IS 6495 Python Programming

Group Project Report

AirAsia Ticketing System

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1. Project Overview

The **AirAsia Ticketing System** is a menu-driven airline management application built with **Python** and **SQLite**. The system is designed to centralize airline operations such as managing airports, aircraft, employees, customers, flight schedules, ticket bookings, and operational reports. The project implements a relational database-backed solution that supports full CRUD functionality, file input through CSV import, user interaction through a command-line menu, dynamic ticket pricing, and business-rule validation for booking operations.

The database structure consists of **7 main tables**: **Airport, Aircraft, Employee, Customer, Flight, Ticket, and FlightCrew**, as shown in the project schema and ERD.

```
============================================================
                  AIRASIA TICKETING SYSTEM
============================================================


------------------------------------------------------------
                         OPERATIONS
------------------------------------------------------------
  1  Airport Management
  2  Aircraft Management
  3  Employee Management
  4  Customer Management
  5  Flight Management
  6  Ticket Management


------------------------------------------------------------
                          REPORTS
------------------------------------------------------------
  7  Flight Sales Report
  8  Employee Flight Report
  9  Passenger Manifest
 10 Flight Crew Report


------------------------------------------------------------
                      SYSTEM UTILITIES
------------------------------------------------------------
 11 Reset Database
 12 Import Sample CSV Data


------------------------------------------------------------
  0  Exit System
------------------------------------------------------------
```

2. Core Features

Full CRUD Implementation

The system supports CRUD operations for the major airline entities:

- **Airport Management**
  Add, search, update, and delete airport records.
- **Aircraft Management**
  Maintain aircraft types and capacities.
- **Employee Management**
  Manage staff records including pilots, cabin crew, mechanics, and ground staff.
- **Customer Management**
  Store and maintain customer information used in ticket booking.
- **Flight Management**
  Create, search, update, and delete flights with associated airport, aircraft, and employee references.
- **Ticket Management**
  Purchase, view, search, update, cancel, and rebook tickets.

Reporting Features

The system also generates operational reports:

- **Flight Sales Report**
- **Employee Flight Assignment Report**
- **Passenger Manifest**
- **Flight Crew Report**

Dynamic Pricing

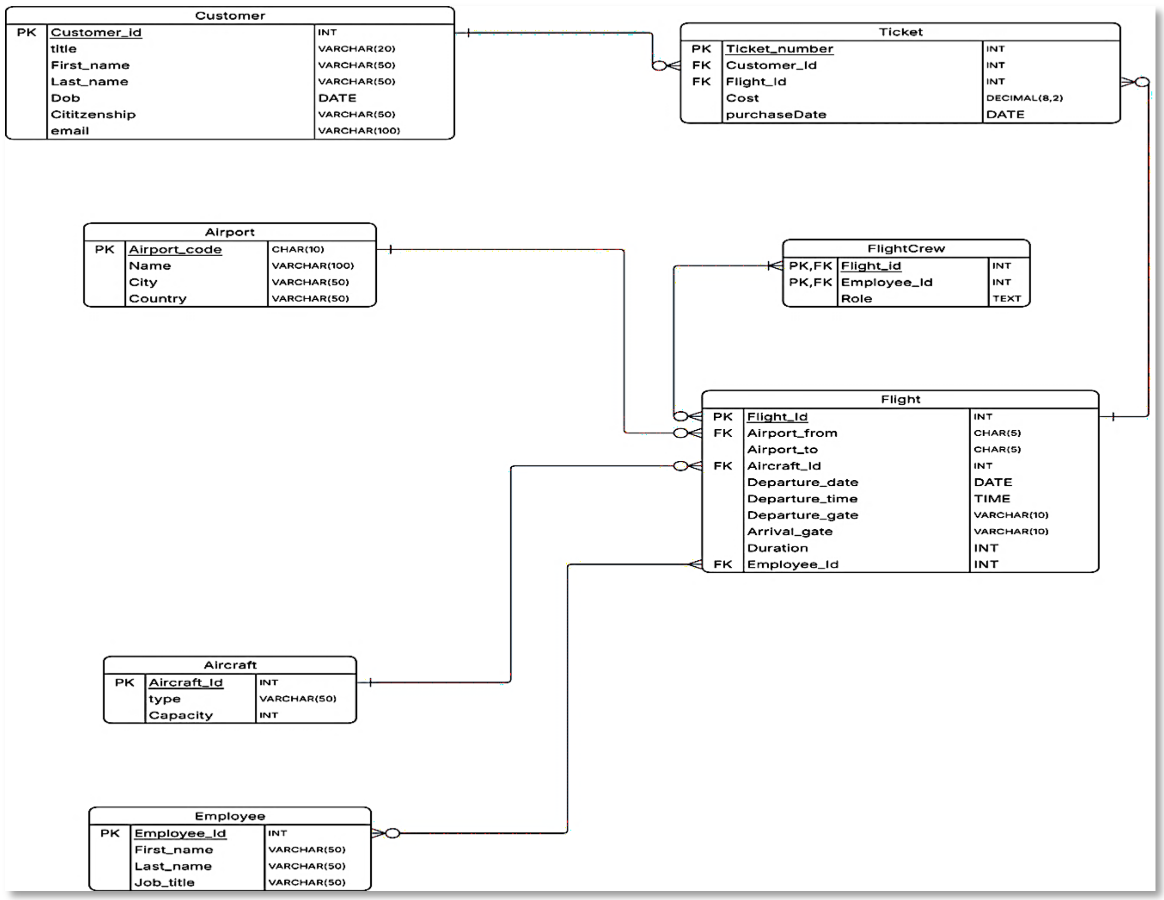
A dynamic pricing engine is integrated into the project to calculate ticket price based on:

- aircraft type
- flight duration
- departure time
- weekend or peak-hour conditions

This supports a more realistic airline pricing model and improves fairness and transparency in booking.

3. Database Design

The AirAsia Ticketing System uses a normalized relational database design in SQLite. The ERD shows how the main entities interact through primary and foreign key relationships.



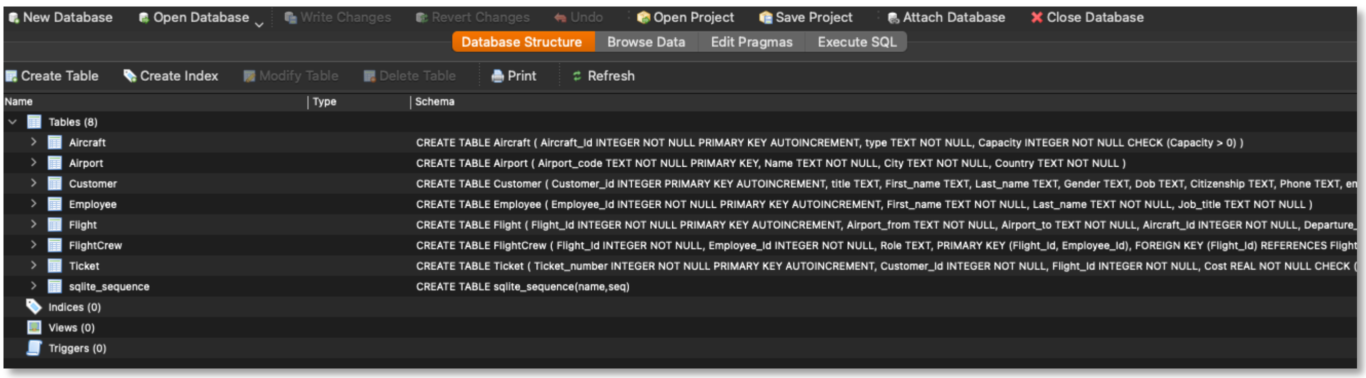
Database Tables

- **Airport**
  Stores airport code, name, city, and country.
- **Aircraft**
  Stores aircraft type and seat capacity.
- **Employee**
  Stores employee first name, last name, and job title.
- **Customer**
  Stores customer profile information.
- **Flight**
  Stores route, aircraft assignment, departure information, duration, and assigned employee.
- **Ticket**
  Stores booking details linked to customer and flight.
- **FlightCrew**
  Junction table connecting flights and employees for crew assignment.

Database Integrity

The database includes several integrity controls:

- Primary keys for all core entities
- Foreign key relationships between dependent tables
- Capacity and duration checks
- Prevention of invalid airport-to-same-airport routes
- Unique customer-flight ticket constraint to prevent duplicate booking for the same flight



4. Object-Oriented Programming Design

The project follows an object-oriented structure where each major entity is represented by a Python class.

Main Classes

- ProjectDB
  Shared database connection layer built on top of db_base.py
- Airport
- Aircraft
- Employee
- Customer
- Flight
- Ticket
- Report
- CSVImporter
- CrewAllocator
- InputHelper
- AirAsiaProject

Design Benefits

This structure improves:

- code readability
- modularity
- maintainability
- easier debugging
- reusability of database logic

Each entity class is responsible for its own CRUD methods, while the AirAsiaProject class acts as the main controller for user interaction.

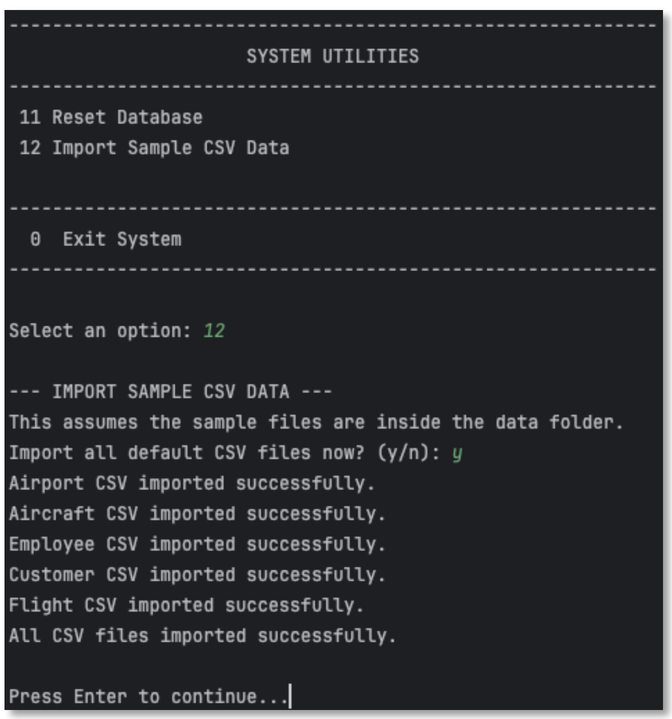
5. File I/O and Data Population

A key requirement of the project was file input/output, and this was implemented through the CSVImporter class.

CSV Files Used

The system imports data from CSV files for:

- airports
- aircraft
- employees
- customers
- flights



Import Process

The importer reads the CSV files using Python's built-in csv.DictReader, validates and cleans values, and inserts them into the database.

This helps quickly initialize the system with realistic sample records for testing and demonstration.

Data Population Benefits

- supports fast setup
- avoids manual entry of all sample records
- demonstrates practical file I/O integration
- aligns with rubric requirements for CSV parsing and persistence

6. Interactive Menu System

The system is designed as a **menu-driven command-line application**.

Main Menu Categories

- Airport Management
- Aircraft Management
- Employee Management
- Customer Management
- Flight Management
- Ticket Management
- Reports
- System Utilities

User Experience

Submenus guide the user through each process step by step, including:

- viewing records
- searching data
- adding or updating records
- booking or cancelling tickets
- generating reports

This makes the system easy to navigate and clearly demonstrates user interaction logic.

7. Booking Workflow

One of the most important parts of the project is the ticket-booking process.

Ticket Purchase Flow

1. Search available flights
2. Select a flight
3. Choose a registered customer
4. Calculate dynamic ticket price
5. Confirm purchase
6. Insert the booking into the Ticket table

```
--- PURCHASE FLIGHT TICKET ---

1 - Search available flights
2 - Select flight and purchase ticket
0 - Return

Select option: 1

Passengers can search by:
City (Bangkok)
Country (Thailand)
Airport name (Changi)
Airport code (SIN)
Origin (city / country / airport / code): BKK
Destination (optional - press Enter to show all routes): DEL
Departure date (MM/DD/YY) optional:
OPT  FLIGHT DATE        TIME   FROM  CITY          TO   CITY          AIRCRAFT          SEATS  DURATION  PRICE
----------------------------------------------------------------------------------------------------------------
1    82     5/29/26     11:25  BKK   Bangkok       DEL  Delhi         Boeing 787-900    290    255       $540.62
2    84     5/30/26     00:30  BKK   Bangkok       DEL  Delhi         Airbus A350neo    325    255       $540.62

Press Enter to continue...
```

```
Press Enter to continue...

--- PURCHASE FLIGHT TICKET ---

1 - Search available flights
2 - Select flight and purchase ticket
0 - Return

Select option: 2
Enter Flight ID to book: 2

OPT  ID   NAME                   EMAIL
-------------------------------------------------------------------
1    1    Amiya Sahoo            amiya@umail.com
2    2    Sarah Lee              sarah.lee@gmail.com
```

```
Select customer option: 1

Ticket price: $162.00
Confirm purchase? (y/n): y
Ticket purchased successfully.
Ticket successfully booked.

Press Enter to continue...
```

Validation During Booking

The system checks:

- whether the flight exists
- whether seats are still available
- whether the customer already has a ticket for that flight

This helps enforce booking rules and prevents invalid transactions.

8. Business Rules Implemented

The system includes business rules that reflect airline ticketing requirements.

Business Rules

- **Direct Flights Only**
  The system supports booking of direct flights only.
- **Individual Bookings**
  Each passenger must have a separate booking.
- **Round Trips as Separate Bookings**
  A round trip must be booked as two independent tickets.
- **Capacity Management**
  Flights cannot be overbooked beyond aircraft seating capacity.
- **Data Integrity**
  Foreign keys and constraints ensure valid relationships among flights, customers, and tickets.
- **Duplicate Ticket Prevention**
  A customer cannot purchase the same flight twice.

```
--- FLIGHT ---

1 - View all flights
2 - Search flights (ID / airport / city)
3 - Add flight
4 - Update flight
5 - Delete flight
0 - Return to main menu

Select option: 1
ID    DATE         DEP TIME   FROM   CITY           TO     CITY           AIRCRAFT TYPE       SEATS  CREW   DEP GATE   ARR GATE   DURATION   PRICE
-------------------------------------------------------------------------------------------------------------------------------------------------
49    5/27/26      07:20      DEL    Delhi          SIN    Singapore      Boeing 787-900      290    16     A12        F6         330        $654.00
53    5/27/26      08:10      BLR    Bangalore      SIN    Singapore      Boeing 787-900      290    14     C6         E3         275        $555.00
57    5/27/26      09:00      BOM    Mumbai         SIN    Singapore      Boeing 777-300ER    390    14     E11        G7         320        $636.00
54    5/27/26      13:50      SIN    Singapore      BLR    Bangalore      Boeing 787-900      290    15     E7         C4         275        $462.50
50    5/27/26      14:10      SIN    Singapore      DEL    Delhi          Boeing 787-900      290    15     C9         A14        330        $545.00
58    5/27/26      16:00      SIN    Singapore      BOM    Mumbai         Boeing 777-300ER    390    15     G6         E12        320        $530.00
```

```
Select customer option: 44

Ticket price: $325.00
Confirm purchase? (y/n): y
An error has occurred while purchasing ticket. This customer already has a ticket for the selected flight.
Ticket successfully booked.

Press Enter to continue...
```

9. Automated Crew Allocation

A unique feature in your project is the CrewAllocator class.

What it Does

When a flight is added, the system automatically assigns crew members based on aircraft capacity.

Crew Categories

- Pilots
- Cabin crew / attendants
- Mechanics
- Ground staff

Benefit

This adds realism to the airline management system and shows that the project goes beyond simple CRUD by including operational logic.

```
--- FLIGHT CREW REPORT ---
Search flights by ID, airport code, or city: 12
OPT  FLIGHT DATE        TIME     FROM   CITY           TO     CITY
---------------------------------------------------------------------------------
1    12     5/28/26     20:20    DPS    Bali           SIN    Singapore
Select flight option: 1

Crew assigned to Flight 12

EMP ID   NAME                      JOB TITLE
------------------------------------------------------------
2        Daniel Tan                Flight Attendant
4        Farid Azmi                Pilot
5        Siti Nur                  Flight Attendant
6        Arun Prakash              Mechanic
8        Linh Tran                 Pilot
9        Maria Santos              Flight Attendant
12       Putri Wijaya              Flight Attendant
17       Kun Joo Cho               Flight Attendant
18       Leilanie Coker            Ground Staff
20       Brian Frerichs            Flight Attendant
23       Astha K C                 Flight Attendant
26       Miles Mccunniff           Flight Attendant
32       Jordan Sharples           Flight Attendant
33       Rob Sheehy                Ground Staff
34       Orion Stonebraker         Pilot
36       Charlotte Wang            Flight Attendant
```

10. Reports and Operational Outputs

The reporting module provides useful summaries for administrative decision-making.

Reports Available

Flight Sales Report

- flight ID
- route
- departure date
- number of tickets sold
- total sales amount

```
--- FLIGHT SALES REPORT ---
FLIGHT  FROM   TO     DATE         TICKETS  TOTAL SALES
-----------------------------------------------------------
22      MNL    BKK    5/29/26      2        650.00
12      DPS    SIN    5/28/26      2        418.60
11      SIN    DPS    5/28/26      2        364.00
34      CGK    SGN    5/30/26      1        212.50
1       SIN    BKK    5/28/26      1        194.40
2       BKK    SIN    5/28/26      1        162.00
3       SIN    BKK    5/28/26      0        0.00
4       BKK    SIN    5/28/26      0        0.00
5       SIN    CGK    5/28/26      0        0.00
```

Employee Flight Report

- employee ID
- employee name
- job title
- number of flights assigned

```
--- EMPLOYEE FLIGHT REPORT ---
EMP ID  EMPLOYEE NAME            JOB TITLE           FLIGHTS
--------------------------------------------------------------
6       Arun Prakash             Mechanic            100
30      Sebastian Perez Parra    Flight Attendant    69
2       Daniel Tan               Flight Attendant    64
12      Putri Wijaya             Flight Attendant    63
9       Maria Santos             Flight Attendant    61
5       Siti Nur                 Flight Attendant    60
14      Whitney Bullock          Flight Attendant    59
```

Passenger Manifest

- ticket number
- customer ID
- passenger name
- email
- flight association

```
Passenger Manifest for Flight 22

TICKET   CUS ID  FIRST NAME      LAST NAME       EMAIL
--------------------------------------------------------------------------------
11       11      Paul            Arnold          paul.arnold@gmail.com
5        12      Cooper          Beck            cooper.beck@gmail.com
12       19      Uday            Gunturu         uday.gunturu@gmail.com
13       20      Justin          Hales           justin.hales@gmail.com
10       45      Daniel          Hill            daniel.hill88@gmail.com
6        44      Olivia          Hill            olivia.hill932@gmail.com

Press Enter to continue...
```

Flight Crew Report

- all employees assigned to a selected flight

```
Crew assigned to Flight 22

EMP ID   NAME                      JOB TITLE
-----------------------------------------------------------
2        Daniel Tan                Flight Attendant
4        Farid Azmi                Pilot
5        Siti Nur                  Flight Attendant
6        Arun Prakash              Mechanic
9        Maria Santos              Flight Attendant
10       Niran Chaiyasit           Ground Staff
16       Corinn Childs             Pilot
17       Kun Joo Cho               Flight Attendant
20       Brian Frerichs            Flight Attendant
23       Astha K C                 Flight Attendant
25       Josh Mcalister            Pilot
26       Miles Mccunniff           Flight Attendant
29       Sina Odejinmi             Flight Attendant
30       Sebastian Perez Parra     Flight Attendant
35       Alina Vannarath           Flight Attendant
37       Jeffrey Zhang             Ground Staff

Press Enter to continue...
```

These reports demonstrate how the database can be used not only for data entry but also for operational analysis.

11. Error Handling and Validation

The project includes exception handling across all main modules.

Error Handling Areas

- database reset failures
- invalid input types
- invalid record lookups
- missing flights or tickets
- duplicate ticket booking
- seat availability errors
- CSV import failures

Validation Features

- non-empty input enforcement
- integer and float validation
- customer and flight existence checks
- seat availability verification before booking

```
Select customer option: 44

Ticket price: $325.00
Confirm purchase? (y/n): y
An error has occurred while purchasing ticket. This customer already has a ticket for the selected flight.
Ticket successfully booked.

Press Enter to continue...
```

This improves reliability and prevents the system from crashing on invalid operations.

12. Ethics and Critical Thinking

The system design also reflects ethical and critical-thinking considerations, which were part of the project rubric and sample documentation structure.

A. Ethical Considerations

Customer Data Protection
Customer data such as name, DOB, email, and citizenship is stored in a structured database and accessed only through defined operations.

Transparent Pricing
The dynamic pricing engine uses visible factors such as duration, aircraft type, peak-hour timing, and weekends. This supports fair and understandable pricing.

No Overbooking
The system checks seat availability before purchase, helping protect customer expectations and operational safety.

Reliable Data Relationships
Foreign key constraints reduce the risk of inconsistent or orphaned records.

B. Critical Thinking in Design

Logical Booking Rules
The project correctly separates customer, ticket, and flight logic into connected but distinct structures.

Scalable Structure

Because the application is modular, it could be extended later with payment systems, login modules, loyalty programs, or web integration.

Operational Realism

The inclusion of crew allocation, reporting, and dynamic pricing shows an effort to simulate a more realistic airline system rather than only a basic database project.

13. Project Strategy and Development Approach

The project was developed in a staged approach similar to the structure used in the sample report.

Phase 1: Analysis and Design

- Studied the airline ticketing requirements
- Identified required entities and relationships
- Built the ERD and schema structure

Phase 2: Infrastructure Development

- Established database connection handling
- Created SQL schema for 7 relational tables
- Implemented CSV import support
- Structured the project using Python classes

Phase 3: Business Logic Implementation

- Added CRUD methods for each entity
- Implemented booking rules and seat checks
- Added reporting features
- Integrated dynamic pricing

Phase 4: User Interface Integration

- Built the interactive menu system
- Connected user options with backend logic
- Added helper methods for cleaner user input and menu flow

14. Files Included in the Project

Based on the files you shared, the project contains the following main components:

- airasia_project.py

  Main application logic and menu system
- create_database.sql

  SQLite schema creation and table definition
- db_base.py

  Base database connection class
- pricing_engine.py

  Dynamic ticket pricing logic
- airasia.sqlite

  SQLite database file
- CSV sample data files
  - airports.csv
  - aircraft.csv
  - employees.csv
  - customers.csv
  - flights.csv
- ERD showing the 7-table structure.

15. Strengths of the Project

This project demonstrates several strengths:

- full CRUD coverage across multiple entities
- relational database design with proper keys
- menu-driven application interface
- CSV-based data loading
- integrated pricing engine
- automated flight crew assignment
- reporting functionality
- object-oriented structure
- practical business-rule enforcement

These features make the system stronger than a simple CRUD demo and show a broader understanding of how Python, SQLite, and structured program design can work together.

16. Suggested Future Improvements

If the project were extended further, possible future improvements could include:

- login and role-based user access
- payment gateway simulation
- email uniqueness constraint for customers
- better date formatting and validation
- web-based or GUI-based interface
- loyalty program or rewards system
- more advanced schedule conflict checking
- dashboard-style reporting
- cascade delete/update strategies where appropriate

17. Conclusion

The **AirAsia Ticketing System** successfully demonstrates the use of Python and SQLite to build a realistic airline database application. The project satisfies the major academic requirements by implementing:

- database CRUD operations
- file I/O through CSV import
- object-oriented program design
- user interaction through menus
- reporting functionality
- error handling and validation
- ethics and critical-thinking considerations

Overall, the project provides a complete and practical example of how database systems, Python programming, and real-world airline business logic can be integrated into one cohesive application.